

O-1 Knapsack Problem using Dynamic Programming

Example: Consider the instance $M=8, n=4$,

$(p_1, p_2, p_3, p_4) = \{1, 2, 5, 6\}$ and

$(w_1, w_2, w_3, w_4) = (2, 3, 4, 5)$.

We may solve the 0/1 dynamic programming using tabulation method

O/1 Knapsack Problem

$m=8$ $P=\{1, 2, 5, 6\}$
 $n=4$ $w=\{2, 3, 4, 5\}$

P_i	w_i	ω	0	1	2	3	4	5	6	7	8
1	2	1	0	0	1	1	1	1	1	1	1
2	3	2	0	0	1	2	2	3	3	3	3
5	4	3	0	0	1	2	5	5	6	7	7
6	5	4	0	0	1	2	5				

$$V[i, \omega] = \max\{V[i-1, \omega], V[i-1, \omega - w_i] + P_i\}$$

$$V[4, 1] = \max\{V[3, 1], V[3, 1-5] + 6\}$$

P_i	w_i	ω	0	1	2	3	4	5	6	7	8
1	2	1	0	0	1	1	1	1	1	1	1
2	3	2	0	0	1	2	2	3	3	3	3
5	4	3	0	0	1	2	5	5	6	7	7
6	5	4	0	0	1	2	5	6			

$$V[i, \omega] = \max\{V[i-1, \omega], V[i-1, \omega - w_i] + P_i\}$$

$$V[4, 5] = \max\{V[3, 5], V[3, 5-5] + 6\}$$

$5, \underline{0+6}$

		ω									
		ω	0	1	2	3	4	5	6	7	8
P_i	ω_i	0	0	0	0	0	0	0	0	0	0
1	2	1	0	0	1	1	1	1	1	1	1
2	3	2	0	0	1	2	2	3	3	3	3
5	4	3	0	0	1	2	5	5	6	7	7
6	5	4	0	0	1	2	5	6	6		

$$V[i, \omega] = \max \{V[i-1, \omega], V[i-1, \omega - \omega_i] + P[i]\}$$

$$V[4, 6] = \max \{V[3, 6], V[3, 6-5] + 6\}$$

$$6, 0+6$$

		ω									
		ω	0	1	2	3	4	5	6	7	8
P_i	ω_i	0	0	0	0	0	0	0	0	0	0
1	2	1	0	0	1	1	1	1	1	1	1
2	3	2	0	0	1	2	2	3	3	3	3
5	4	3	0	0	1	2	5	5	6	7	7
6	5	4	0	0	1	2	5	6	6	7	

$$V[i, \omega] = \max \{V[i-1, \omega], V[i-1, \omega - \omega_i] + P[i]\}$$

$$V[4, 7] = \max \{V[3, 7], V[3, 7-5] + 6\}$$

$$7, 1+6$$

		ω									
		ω	0	1	2	3	4	5	6	7	8
P_i	ω_i	0	0	0	0	0	0	0	0	0	0
1	2	1	0	0	1	1	1	1	1	1	1
2	3	2	0	0	1	2	2	3	3	3	3
5	4	3	0	0	1	2	5	5	6	7	7
6	5	4	0	0	1	2	5	6	6	7	8

$$V[i, \omega] = \max \{V[i-1, \omega], V[i-1, \omega - \omega_i] + P[i]\}$$

$$V[4, 8] = \max \{V[3, 8], V[3, 8-5] + 6\}$$

$$7, 2+6$$

Hint for filling the Table : (last row is given as example)

Once the table is filled we should start with the last row to consider whether object x4 should be included in the solution or not. Since the value 8 is the maximum profit observed in the object x4 row, we should start from that one. Now to Check whether 8 is present in the previous row (object x3)or not. It is not, ie. Because of this object (x4) alone we get the maximum profit. Now we should perform the operation [sack capacity – the profit gained] as $[8-6]=2$.

Next step is to check with the remaining profit 2. To check whether object x3 should be included in the solution or not thus by verifying value 2 is present in the previous row(object x2). But in this case 2 is present in the previous row too. It's clear that object x3 alone not giving the profit. Hence we may not include the object x3 in the solution.

Next is to check the profit 2 is present in the object x2 row) or not. Yes it is present, but its previous row (object x1) didn't have the value 2. Hence it is clear that because of the object 2 alone we get the profit. Now we should include object X2 in the solution thus by performing the operation [sack capacity – the profit gained] as $[2-2]=0$.

Now the sack is full, but we should ensure and check whether the first object should be included in the solution or not. It is clear that the value 0 is present in the first object row x1 as well as its previous row too. Hence we cannot include object x1 it in the solution .

Therefore , we get the solution as $\{ x1,x2,x3,x4 \} = \{0,1,0,1\}$

(P, W)

① $(8, 8) \in S^4$
 but $(8, 8) \notin S^3 \therefore x_4 = 1$
 $(8-6, 8-5) = (2, 3)$

② $(2, 3) \in S^3$
 and $(2, 3) \in S^2 \therefore x_3 = 0$

③ $(2, 3) \in S^2$
 but $(2, 3) \notin S^1 \therefore x_2 = 1$
 $(2-2, 3-3) = (0, 0)$

④ $(0, 0) \in S^1$ and $(0, 0) \in S^0 \therefore x_1 = 0$

$S^0 = \{(0, 0)\}$
 $S^1 = \{(1, 2)\}$
 $S^2 = \{(0, 0), (1, 2)\}$
 $S^3 = \{(2, 3), (3, 5)\}$
 $S^4 = \{(0, 0), (1, 2), (2, 3), (3, 5)\}$
 $S^5 = \{(5, 4), (6, 6), (7, 7), (8, 9)\}$
 $S^6 = \{(0, 0), (1, 2), (2, 3), (3, 5), (5, 4), (6, 6), (7, 7)\}$
 $S^7 = \{(6, 5), (7, 7), (8, 8), (12, 11), (13, 12)\}$
 $S^8 = \{(0, 0), (1, 2), (2, 3), (5, 4), (6, 6), (7, 7), (8, 8)\}$

Hence the solutions $\{x_1, x_2, x_3, x_4\} = \{0, 1, 0, 1\}$